ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Section A: objective questions (30) · Section B: robot configurations and coordinate systems, kinematics, motors and drives, sensors and feedback, mobile robots and robot programming methods (70).

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INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A robot with three linear (prismatic) axes along X, Y and Z is a …

A SCARA robot. B articulated robot.
C Cartesian (gantry) robot. D spherical robot.

1.1.2 A robot arm with two parallel rotary joints that is very fast in horizontal pick-and-place work is a …

A SCARA robot. B Cartesian robot. C delta robot only. D mobile robot.

1.1.3 Calculating the position of the end effector from known joint angles is …

A inverse kinematics. B forward kinematics.
C dynamics. D teaching.

1.1.4 A stepper motor with a step angle of 1,8° needs … steps for one full revolution.

A 180 B 360 C 100 D 200

1.1.5 Changing the average voltage to a DC motor by switching it on and off quickly is called …

A pulse-width modulation (PWM). B regenerative braking.
C commutation. D rectification.

1.1.6 Which circuit allows a DC motor to run in both directions?

A Voltage divider B Rectifier C H-bridge D Low-pass filter

1.1.7 Programming a robot by moving it to each position with a hand-held pendant and saving the points is …

A offline simulation. B machine learning.
C teach-pendant programming. D block programming only.

1.1.8 A sensor that tells the controller the exact angle of a robot joint is a/an …

A rotary encoder. B LDR.
C microphone. D thermocouple.

1.1.9 A mobile robot with two driven wheels on one axle that steers by changing their speeds uses …

A Ackermann steering. B tracks.
C legs. D differential drive.

1.1.10 The accuracy of a robot is how close it gets to …

A its previous position. B the commanded (target) position.
C its maximum speed. D its payload.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 An articulated robot usually has six degrees of freedom.

1.2.2 Inverse kinematics calculates the joint angles needed to reach a given position.

1.2.3 A servo motor contains a feedback device to control its position.

1.2.4 A 50% PWM duty cycle on a 12 V supply gives an average of 12 V.

1.2.5 Gearboxes on robot joints increase torque and reduce speed.

1.2.6 The world coordinate system is fixed to the robot's base or work cell.

1.2.7 Infrared sensors are not affected by sunlight.

1.2.8 A gyroscope measures rotation (angular velocity).

1.2.9 Stepper motors always need an encoder to move to a position.

1.2.10 Offline programming lets engineers program a robot in a simulation without stopping production.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Joint that slides in a straight line	A	prismatic joint
1.3.2	Joint that rotates about an axis	B	IMU
1.3.3	Coordinate system fixed to the robot's tool centre point	C	LiDAR
1.3.4	Percentage of time a PWM signal is ON	D	linear interpolation
1.3.5	Sensor combining an accelerometer and gyroscope	E	delta robot
1.3.6	Laser scanner that maps the surroundings	F	revolute joint
1.3.7	Estimating position from wheel rotations	G	tool frame
1.3.8	Gripper that holds objects using suction	H	odometry

1.3.9	Motion where the tool moves in a straight line between two points	I	duty cycle
1.3.10	Robot built from parallel arms, used for very fast picking	J	vacuum gripper

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: CONFIGURATIONS AND COORDINATE SYSTEMS

2.1 Describe the joints (linear or rotary) and the shape of the work envelope of EACH configuration: (a) Cartesian; (b) cylindrical; (c) articulated. (6)

2.2 Recommend a robot configuration for EACH task and give a reason: (a) placing chocolates into boxes on a fast conveyor; (b) welding a car body. (4)

2.3 Name and explain THREE coordinate systems (frames) used to program an industrial robot. (6)

2.4 Explain the difference between accuracy and repeatability. (4)

[20]

QUESTION 3: KINEMATICS AND MOTION

3.1 A two-link planar robot arm has link lengths L_1 = 30 cm and L_2 = 20 cm. Joint 1 is at θ_1 = 30° from the X-axis and joint 2 is at θ_2 = 60° relative to link 1. Use $x = L_1\cos\theta_1 + L_2\cos(\theta_1 + \theta_2)$ and $y = L_1\sin\theta_1 + L_2\sin(\theta_1 + \theta_2)$.

3.1.1 Calculate the x and y coordinates of the end effector. (6)

3.1.2 What is the maximum reach of this arm? (1)

3.2 Explain why inverse kinematics is more difficult than forward kinematics. (2)

3.3 Describe the difference between point-to-point motion and continuous-path motion, and give a task for each. (4)

[13]

QUESTION 4: MOTORS, DRIVES AND SENSORS

4.1 A 12 V DC motor is controlled with PWM. Calculate the average voltage at a duty cycle of 75%. (2)

4.2 A stepper motor has a step angle of 1,8° and drives a lead screw with a pitch of 2 mm. Calculate: (a) the number of steps to turn the screw through 90°; (b) the distance the nut moves for 1 000 steps. (5)

4.3 Compare DC motors, servo motors and stepper motors under the headings control of position and typical robot use. (6)

4.4 Explain how a rotary encoder gives feedback to a robot controller. (3)

4.5 Explain how an ultrasonic sensor calculates distance. A sound pulse returns after 2,9 ms; calculate the distance if sound travels at 343 m/s. (4)

[20]

QUESTION 5: MOBILE ROBOTS AND PROGRAMMING

5.1 Explain how a differential-drive robot moves forward, turns left and spins on the spot. (3)

5.2 Name THREE methods of programming a robot and give ONE advantage of each. (6)

5.3 A delivery robot must navigate a hospital corridor. Name FOUR sensors it could use and the purpose of each. (4)

5.4 Name FOUR types of end effectors and a task for each. (4)

[17]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER